

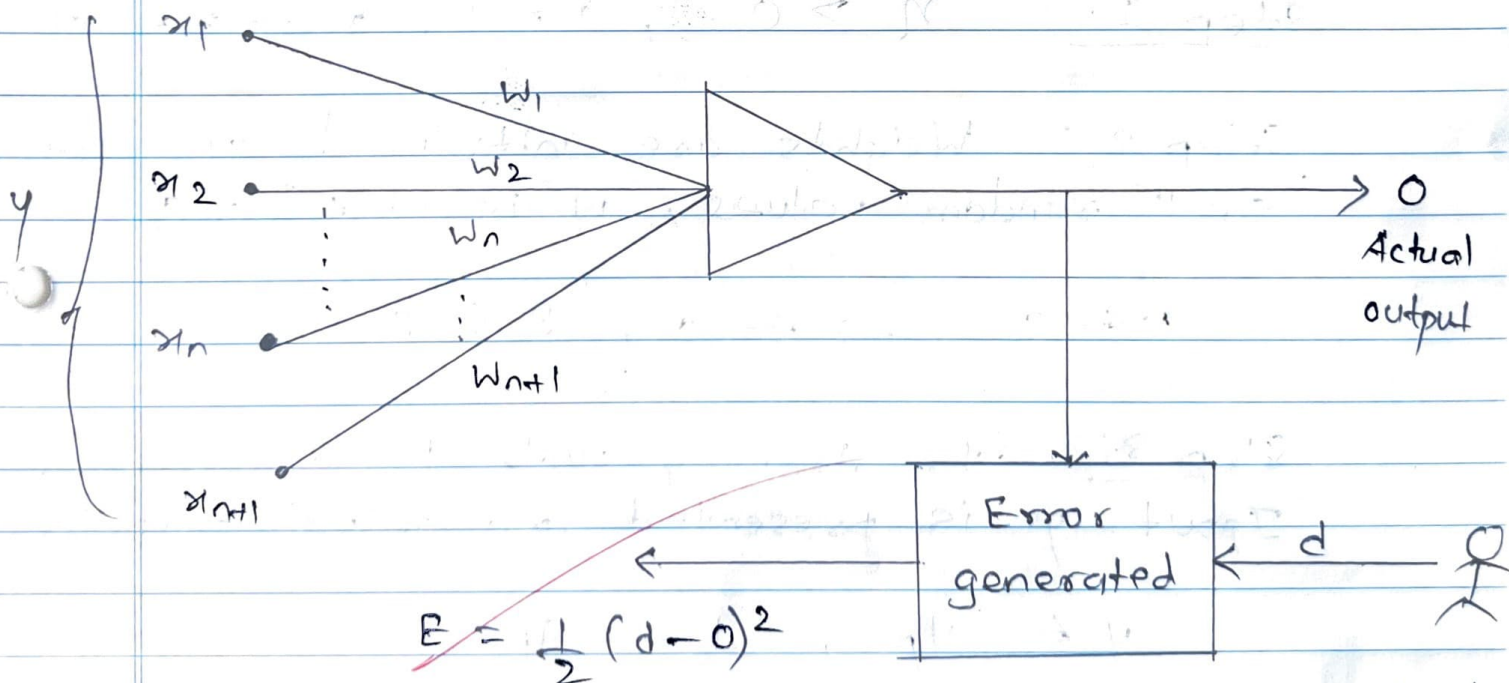
Assignment-2

Topic - Continuous Perceptron Learning Algorithm

Q.1 Explain continuous perceptron learning rule with the help of neat Feed Forward neural network architecture diagram.

⇒

- The perceptron learning algorithm is a fundamental supervised learning technique used for binary classification.
- In its continuous form, the perceptron updates weights using a continuous activation function rather than a step function.



Note:

$$y_i = x_i, \quad i = 1, 2, \dots, n$$

Teacher's
desired
output

→ Working :

Given are P pairs,

$$\{x_1, d_1, x_2, d_2, \dots, x_p, d_p\}$$

$$x_i = (n \times 1)$$

$$d_i = (1 \times 1)$$

$$i = 1, 2, \dots, p$$

Augmented input vectors are used:

$$y_i = \begin{bmatrix} x_i \\ 1 \end{bmatrix}, \text{ for } i = 1, 2, \dots, p$$

Step 1 : $\eta > 0$, $\lambda = 1$, $E_{max} = 0$ chosen

Step 2 : Weights are initialized at w at small random values, w is $(n+1) \times 1$

$$K \leftarrow 1, p \leftarrow 1, E \leftarrow 0$$

Step 3 : The training cycle begins here.
Input y is presented and output computed.

$$y \leftarrow y_p, d \leftarrow d_p$$

$$o \leftarrow \text{sgn}(wt \cdot y)$$

Step 4: Weights are updated

$$w \leftarrow w + \frac{1}{2} \eta (d-o) (1-o^2) y$$

Step 5: Cycle error is computed

$$E \leftarrow \frac{1}{2} (d-o)^2 + E$$

Step 6: IF $p < P$, then $p \leftarrow p+1$,
 $k \leftarrow k+1$ and goto step 3, otherwise
goto step 7.

Step 7: The training cycle is completed.

For $E < E_{max}$ terminate the training session. Output weights, k and E .

IF $E > E_{max}$, then $E \leftarrow 0$, $p \leftarrow 1$,
and enter the new training cycle
by going to step 3.

Q.2 Apply continuous perceptron learning rule to obtain weights after 1 iteration for the following data patterns. Assume learning rate $C = 0.1$. Use bipolar continuous activation function.

$$X_1 = [1 \ -2 \ 0 \ 1], \quad X_2 = [0 \ 1.5 \ -0.5 \ -1]$$

$$X_3 = [-1 \ 1 \ 0.5 \ -1],$$

$$d_1 = -1, \quad d_2 = -1, \quad d_3 = 1$$

$\Rightarrow$ Solⁿ:-

$$W_1^T = [1 \ -1 \ 0 \ 0.5]$$

For delta learning with bipolar continuous activation function.

$$net_i = W_i^T X$$

$$O_i = f(net_i) = \frac{2}{1 + e^{(-net_i)}} - 1$$

$$f'(net_i) = \frac{1}{2} (1 - O_i^2)$$

$$\Delta W_i = C (d - O_i) f'(net_i) X$$

Step 1: $X = X_1$, $d = d_1$

$$net_1 = W_1^t X$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix} = 3.5$$

$$\begin{aligned} O_1 &= f(net_1) = \frac{2}{1 + e^{(-net)}} - 1 \\ &= \frac{2}{1 + e^{-3.5}} - 1 \\ &= 0.94 \end{aligned}$$

$$\begin{aligned} f'(net_1) &= \frac{1}{2} (1 - O^2) \\ &= 0.037 \end{aligned}$$

$$\Delta W_1 = f'(net_1) \times c (d - O_1) \times X_1$$

$$= 0.037 \times 0.1 (-1 - 0.94) \times \begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -0.011 \\ 0.022 \\ 0 \\ -0.011 \end{bmatrix}$$

$$\Delta W_2 = W_1 + \Delta W = \begin{bmatrix} 0.99 \\ -0.98 \\ 0 \\ 0.489 \end{bmatrix}$$

Step 2: $x = x_2$, $d = d_2$

$$\text{net}_2 = W_2^t x_2$$

$$= \begin{bmatrix} 0.99 & -0.98 & 0 & 0.489 \end{bmatrix} \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$$

$$= -1.959$$

$$f'(\text{net}_2) = \frac{2}{1 + e^{-1.959}} - 1 = -0.753$$

$$f'(\text{net}_2) = \frac{1}{2} (1 - 0^2) = 0.217$$

$$\Delta W = 0.1 \times 0.217 \times (-1 + 0.753) \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$$

$$\Delta W = \begin{bmatrix} 0 \\ -0.0027 \\ 0.0027 \\ 0.0053 \end{bmatrix}$$

$$W_3 = \Delta W + W_2$$

$$= \begin{bmatrix} 0.99 \\ -0.973 \\ 0.027 \\ 0.489 \end{bmatrix}$$

Step 3: $X = X_3, d = d_3$

$$net_3 = W_3^T X_3$$

$$= \begin{bmatrix} 0.99 & -0.973 & 0.027 & 0.489 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$= -2.44$$

$$O_3 = f(net_3) = \frac{2}{1 + e^{+2.44}} - 1 = -0.84$$

$$\Delta W_2 = 0.1 \times 0.147 \times (1 + 0.84) \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$\Delta W = \begin{bmatrix} -0.027 \\ 0.027 \\ 0.018 \\ 0.027 \end{bmatrix}$$

$$W_4 = W_3 + \Delta W$$

$$= \begin{bmatrix} 0.963 \\ -0.946 \\ 0.016 \\ 0.516 \end{bmatrix}$$

~~(B) 9/14/25~~